Ch 9.2: Support Vector Classifier

Lecture 28 - CMSE 381

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Wed, Nov 15, 2023

Announcements

Last time:

- 9.1 Maximal Margin Classifier

This lecture:

- 9.2 Support Vector Classifier

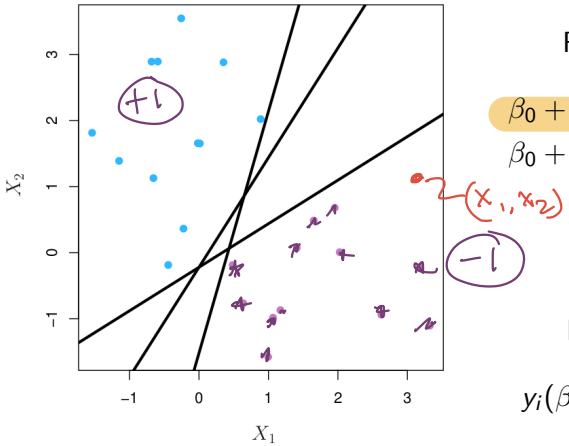
Announcements:

- HW #7.... Mon Nov 20

Section 1

Last time

Separating Hyperplane



Require that for every data point:

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} > 0 \text{ if } y_i = 1$$
$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} < 0 \text{ if } y_i = -1$$

Equivalently

Require that for every data point

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) > 0$$

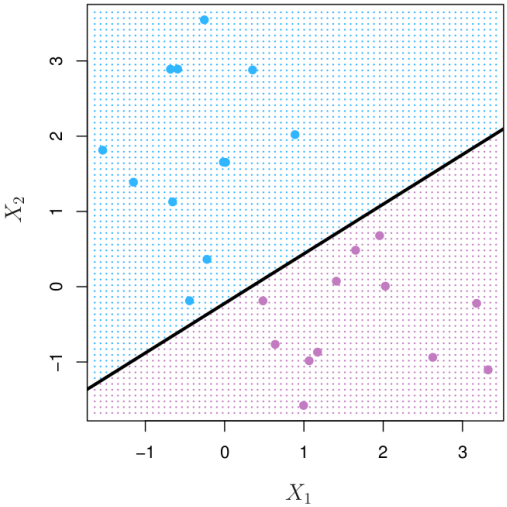
Separating hyperplane becomes a classifier

If you have a separating hyperplane:

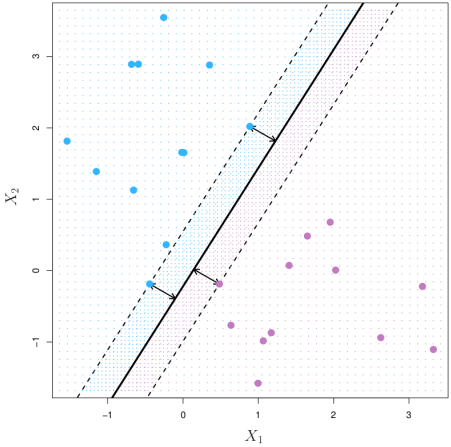
- Check

$$f(x^*) = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \cdots + \beta_p x_p^*$$

- If positive, assign $\hat{y} = 1$
- If negative, assign $\hat{y} = -1$



Maximal margin classifier



- For a hyperplane, the *margin* is the smallest distance from any data point to the hyperplane.
- Observations that are closest are called *support vectors*.
- The *maximal margin hyperplane* is the hyperplane with the largest margin
- The classifier built from this hyperplane is the *maximal margin classifier*.

Mathematical Formulation

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \; M$$

margin want this to be big!

$$\text{subject to} \sum_{j=1}^{p} \beta_j^2 = 1,$$

can get this by dividing by $\sum_{j=1}^{p} \beta_j^2$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M \;\; \forall\, i = 1, \ldots, n$$

requires this!

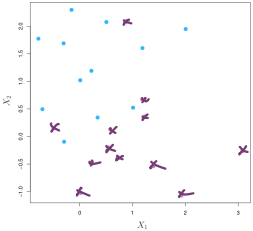
abs val = distance from data pt to the line

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p = 0$$

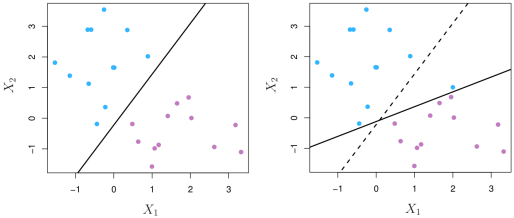
requires that every data pt is $\geq M$ from hyperplane

Problems

Might be no separating hyperplane



Sensitivity to new points



Section 2

Support Vector Classifier

according to the book...

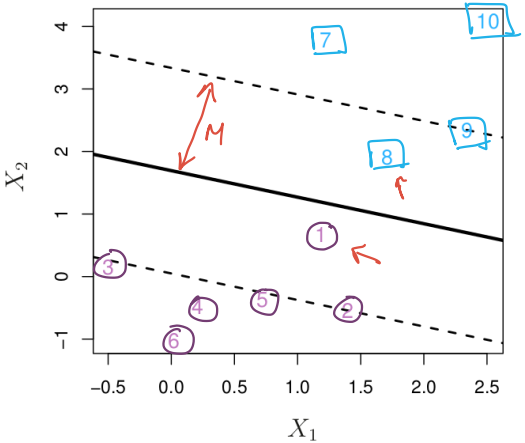
Basic idea

- Be ok with a classifier that isn't perfect on the training data
- Aim for robustness to individual observations
- Better classification on most of training observations

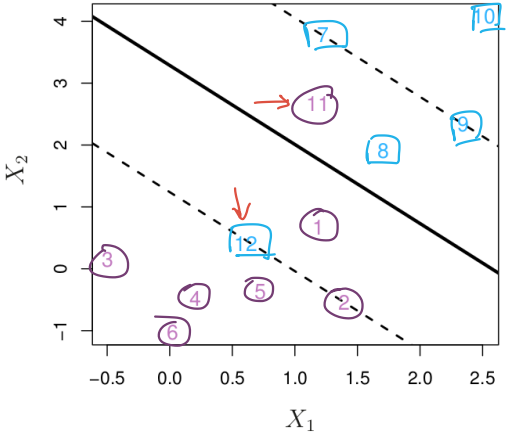
↳ Result: Support vector classifier

"Soft margins"

Soft margin



Some data on the wrong side margin (but correctly classified)



Some points on the wrong side of the hyperplane

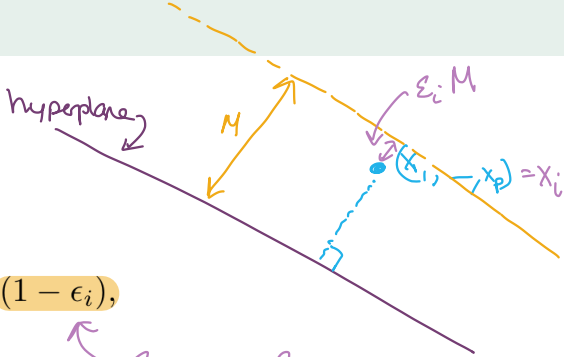
Mathematical Formulation of SVC

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$\underbrace{y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip})}_{\text{distance to the hyperplane}} \geq M(1-\epsilon_i),$$

$$\underbrace{\epsilon_i \geq 0,}_{} \quad \underbrace{\sum_{i=1}^{n} \epsilon_i \leq C,}_{\text{Budget for error}}$$



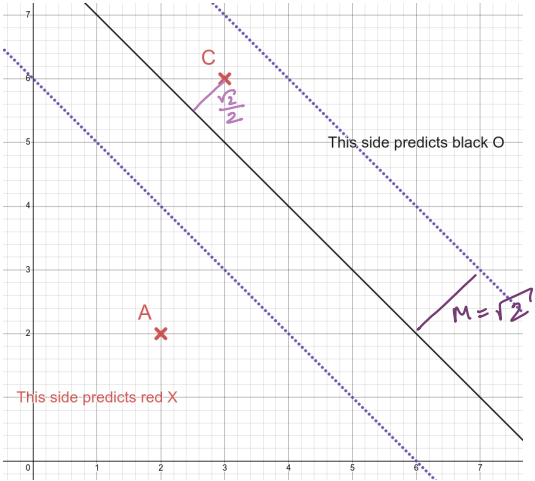
ϵ_i cost for each data pt (x_i, y_i)

Hyper parameter $C \geq 0$

$\hookrightarrow C = 0 \Rightarrow$ max'l margin hyperplane

Find positive ε's that will satisfy this

Fix $M = \sqrt{2}$ $\qquad y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \varepsilon_i)$



Data point A:

Correctly classified, correct side of margin

distance from A to hyperplane $\geq M$

Set $\varepsilon_i = 0$

Data point C:

Wrong side of hyperplane

Left side = neg — dist to the hyperplane

dist $= \frac{\sqrt{2}}{2}$ $\qquad$ Want ε_i s.t.

LHS $= -\frac{\sqrt{2}}{2}$ $\qquad -\frac{\sqrt{2}}{2} \geq \sqrt{2}(1 - \varepsilon_i)$

$\varepsilon_i \geq 1.5$ $\qquad -\frac{1}{2} \geq 1 - \varepsilon_i$

$\varepsilon_i \geq 1.5$

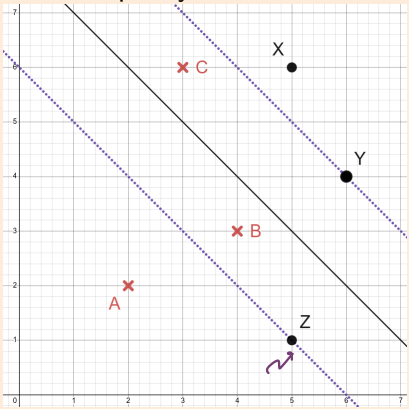
What is ε?

$\frac{\sqrt{2}}{2} \geq \sqrt{2}(1-\varepsilon_i)$

$\frac{1}{2} \geq 1-\varepsilon_i \quad \varepsilon_i \geq \frac{1}{2}$

Fix $M = \sqrt{2}$ $\qquad y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1-\varepsilon_i)$

Fill in the table so that the inequality is satisfied.



Point	Left Side	ε_i	$M(1-\varepsilon_i)$
A	$2\sqrt{2}$	0	$\sqrt{2}$
B	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	$\sqrt{2}(1-\frac{1}{2}) = \frac{\sqrt{2}}{2}$
C	$-\frac{\sqrt{2}}{2}$	1.5	$-\frac{\sqrt{2}}{2}$
X	$\frac{3\sqrt{2}}{2}$	0	$\sqrt{2}$
Y	$\sqrt{2}$	0	$\sqrt{2}$
Z	$-\sqrt{2}$	2	$\sqrt{2}(1-2) = -\sqrt{2}$

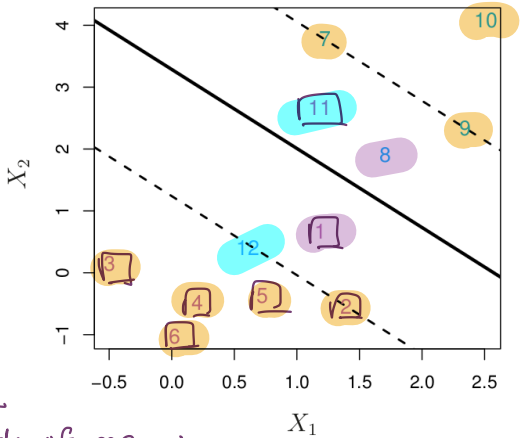
What is ε?

If $\varepsilon_i = 0$ $\Rightarrow$ pt is on the correct side of hyperplane —and— correct side of the margin

If $\varepsilon_i > 0$ $\Rightarrow$ the wrong side of the margin

If $\varepsilon_i \in [0,1]$ $\Rightarrow$ the right side of hyperplane but wrong side of margin

If $\varepsilon_i > 1$ $\Rightarrow$ the wrong side of hyperplane



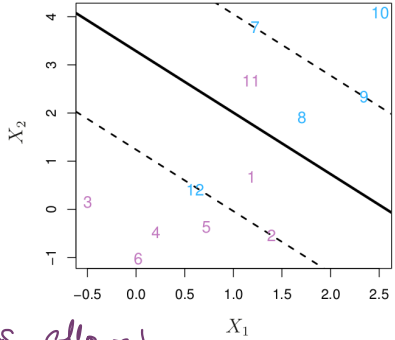
What is C?

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1-\epsilon_i),$$

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

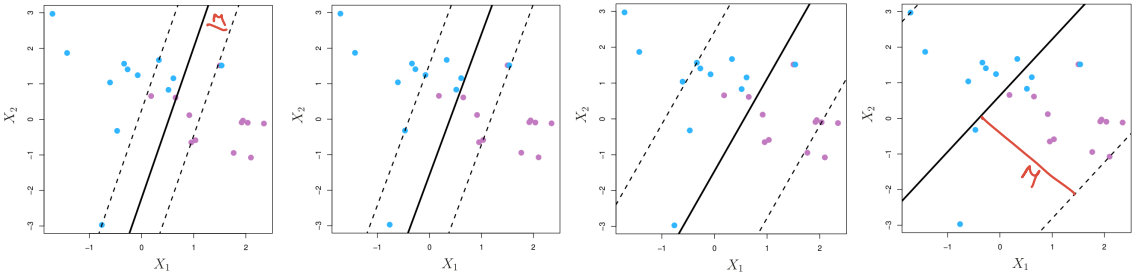


↳ $C=0$ means no violations allowed

$C>0$ means at most C points misclassified (but could have mix of misclassified and violating margin)

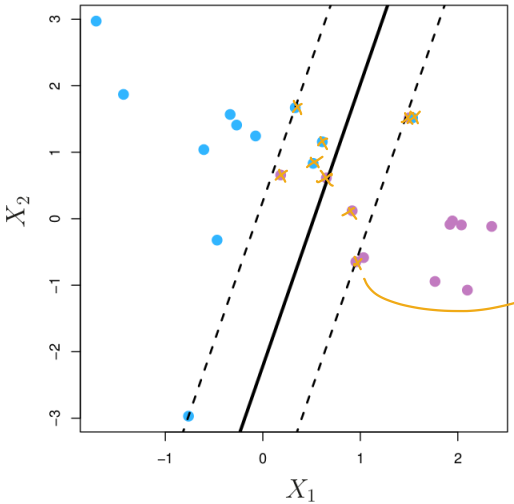
Sum from example was $C=4$

Examples messing with C



Increasing $C \rightarrow$

What affects the hyperplane?



Observations on or violating margin (wrong side of margin)

are support vectors

↳ the ones w/ $\varepsilon_i > 0$

Support vectors

Coding

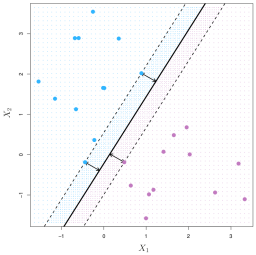
TL;DR

Maximal Margin Classifier

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M \quad \forall\, i = 1, \ldots, n$$



Support Vector Classifier

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\, M}{\text{maximize}} \quad M$$

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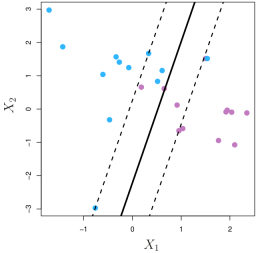
$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$



Next time

Status	Lec #	Date			Reading	Homeworks
		Mon	Oct 23	No class - Fall break		
		Wed	Oct 25	**Midterm #2**		
Done	20	Fri	Oct 27	Dimension Reduction	6.3	
Done	21	Mon	Oct 30	More dimension reduction; High dimensions	6.4	
Done	22	Wed	Nov 1	Polynomial & Step Functions	7.1,7.2	
Pushed	23	Fri	Nov 3	Step Functions; Basis functions; Start Splines	7.2 - 7.4	
	24	Mon	Nov 6	Regression Splines	7.4	~~HW #6 Due~~
	25	Wed	Nov 8	Decision Trees	8.1	HW #6 Due
	26	Fri	Nov 10	Random Forests	8.2.1, 8.2.2	
	27	Mon	Nov 13	Maximal Margin Classifier	9.1	
	28	Wed	Nov 15	SVC	9.2	
	29	Fri	Nov 17	SVM	9.3, 9.4	
	30	Mon	Nov 20	Single layer NN	10.1	HW 7
	31	Wed	Nov 22	Virtual: Project office hours		
		Fri	Nov 24	No class - Thanksgiving		
		Mon	Nov 27	**Review**		
		Wed	Nov 29	**Midterm #3**		
	32	Fri	Dec 1	Multi Layer NN	10.2	
	33	Mon	Dec 4	CNN	10.3	
	34	Wed	Dec 6	Unsupervised Learning & Clustering	12.1, 12.4	
	35	Fri	Dec 8	Virtual: Project office hours		Project due